

SQL DATA ANALYSIS REPORT

MEDICARE HEALTHCARE OPERATIONS & REVENUE ANALYTICS

Prepared By:

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Technology Used:

MySQL

Dataset:

Medicare Dataset

2. PROJECT OVERVIEW

MediCare Healthcare Operations & Revenue Analytics is a MySQL-based healthcare analytics project developed to analyze hospital, patient, appointment, admission, treatment, laboratory, pharmacy, billing and payment information.

2.2 Business Objective

The objective of the project is to create a structured relational database and use SQL-based analysis to evaluate healthcare operations, patient activity, service utilization, billing performance and payment collection.

2.3 Project Scope

The project focuses on healthcare operational and financial analysis using relational database concepts and SQL queries.

3. BUSINESS REQUIREMENTS

Stakeholders

- Hospital Management
- Hospital Administrators
- Finance and Billing Team
- Operations Team
- Department Heads
- Data / BI Analysts

Key Business Requirements

- Analyze hospital activity
- Analyze department workload
- Analyze doctor workload
- Analyze patient activity
- Analyze admissions and length of stay
- Analyze treatment utilization
- Analyze laboratory activity
- Analyze pharmacy activity
- Analyze billing and payment collection
- Calculate important healthcare KPIs
- Generate business insights and recommendations

4. DATASET & DATABASE

Database Name

Database: medicare

Tables

The database contains the following tables:

1. Hospitals
2. Departments
3. Doctors
4. Patients
5. Rooms
6. Appointments
7. Admissions
8. Treatments
9. Insurance
10. Medicines
11. Pharmacy
12. Laboratory
13. Employees
14. Billing
15. Payments

Data Loading

The source data was imported into the respective MySQL tables using the MySQL Data Import Wizard.

5. DATABASE DESIGN

Database Relationships

The database follows a relational healthcare structure connecting patients, appointments, admissions, healthcare services, billing and payments.

Healthcare Data Flow

Patient → Appointment → Admission → Treatment / Laboratory / Pharmacy → Billing
→ Payment

Primary Keys and Foreign Keys

Primary keys were defined to uniquely identify records in each table. Foreign keys were used to establish relationships between related healthcare entities and maintain referential integrity.

6. DATA VALIDATION

Validation Performed

The following validation checks were performed:

- Record count validation
- Primary-key duplicate validation
- NULL-value investigation
- Foreign-key integrity validation
- Numeric-value validation
- Date validation

Data Quality Decision

The identified NULL and unusual values were investigated without making unsupported changes to the original source records. No unnecessary data replacement was performed.

7. HOSPITAL & DEPARTMENT ANALYSIS

Hospital Activity

1.Business Question:

Which hospitals have the highest operational activity?

SQL Query:

```
SELECT
    h.hospital_id,
    h.hospital_name,
    COUNT(DISTINCT a.appointment_id) AS total_appointments,
    COUNT(DISTINCT ad.admission_id) AS total_admissions,
    COUNT(DISTINCT a.patient_id) AS unique_patients
FROM Hospitals h
LEFT JOIN Appointments a
    ON h.hospital_id = a.hospital_id
LEFT JOIN Admissions ad
    ON h.hospital_id = ad.hospital_id
GROUP BY h.hospital_id, h.hospital_name
ORDER BY total_admissions DESC, total_appointments DESC;
```

Result:

hospital_id	hospital_name	total_appointments	total_admissions	unique_patients
H005	MediCare Belagavi North Campus	493	167	434
H004	MediCare Tiruchirappalli North Campus	166	69	155
H010	MediCare Thrissur Heights	154	69	144
H023	MediCare Erode Heights 2	156	67	148
H019	MediCare Vijayawada South Campus 2	160	66	151
H013	MediCare Chennai Institute of Medical Sciences	165	65	160
H003	MediCare Vellore Annex	166	63	157
H007	MediCare Kozhikode North Campus	168	62	155
H026	MediCare Thiruvananthapuram Superspeciality ...	171	61	161
H008	MediCare Vellore Multispeciality Centre 2	165	60	156
H002	MediCare Coimbatore Central	172	59	162
H014	MediCare Tirunelveli Central	166	59	156
H027	MediCare Thiruvananthapuram Multispeciality C...	164	59	158
H016	MediCare Visakhapatnam Heights	151	59	140
H001	MediCare Kochi North Campus	186	58	172
H029	MediCare Hubballi North Campus	166	58	158
H021	MediCare Vellore Superspeciality Centre 4	161	58	154
H009	MediCare Erode Annex	163	57	151
H030	MediCare Thiruvananthapuram Institute of Medi...	152	56	141
H018	MediCare Kozhikode Central 2	156	55	151
H022	MediCare Warangal North Campus	152	54	137
H024	MediCare Thiruvananthapuram Annex 2	174	53	160
H006	MediCare Madurai Superspeciality Centre	171	52	161
H017	MediCare Vellore Superspeciality Centre 3	158	52	148
H028	MediCare Madurai Central 2	169	50	157
H012	MediCare Vijayawada Central	159	46	151
H025	MediCare Coimbatore Annex 2	132	45	127
H015	MediCare Guntur North Campus	180	42	171
H011	MediCare Salem South Campus	165	42	159
H020	MediCare Thiruvananthapuram Heights	39	13	37

Business Insight:

Medicare shows the highest operational activity based on its appointment and admission volume. This indicates a comparatively higher level of patient service demand at the hospital.

Recommendation:

Management can monitor the workload of **Medicare** and ensure that staff, rooms and other resources are appropriately planned to support the higher activity level.

Department Workload

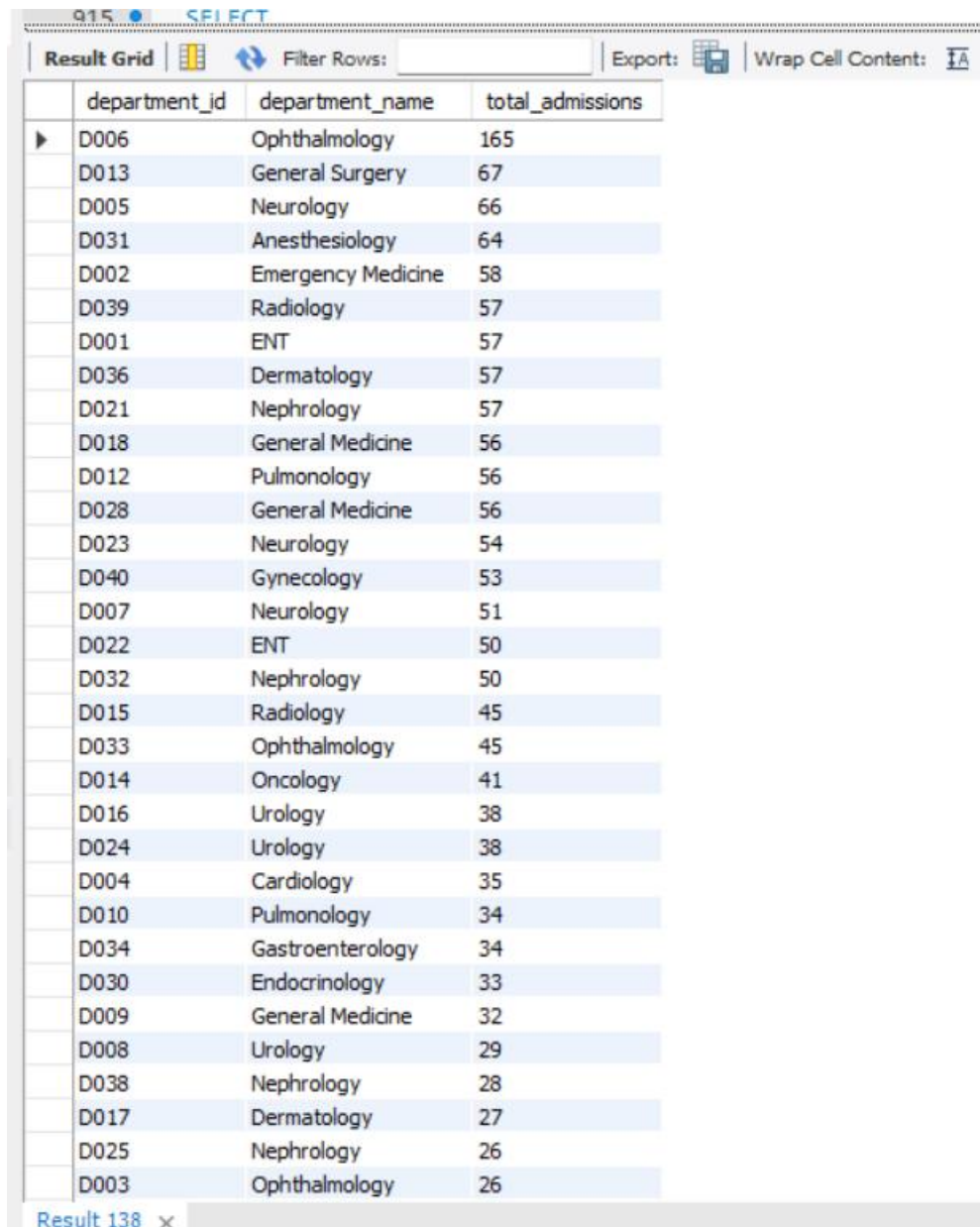
2.Business Question:

Which departments have the highest admission workload?

SQL Query:

```
SELECT
    d.department_id,
    d.department_name,
    COUNT(DISTINCT ad.admission_id) AS total_admissions
FROM Departments d
LEFT JOIN Admissions ad
    ON d.department_id = ad.department_id
GROUP BY d.department_id, d.department_name
ORDER BY total_admissions DESC;
```

Result:



The screenshot shows a SQL query result grid with the following columns: department_id, department_name, and total_admissions. The results are sorted in descending order of total_admissions. The top row is D006 (Ophthalmology) with 165 admissions. The bottom row is D003 (Ophthalmology) with 26 admissions. The grid includes a toolbar with options like 'Filter Rows', 'Export', and 'Wrap Cell Content'.

department_id	department_name	total_admissions
D006	Ophthalmology	165
D013	General Surgery	67
D005	Neurology	66
D031	Anesthesiology	64
D002	Emergency Medicine	58
D039	Radiology	57
D001	ENT	57
D036	Dermatology	57
D021	Nephrology	57
D018	General Medicine	56
D012	Pulmonology	56
D028	General Medicine	56
D023	Neurology	54
D040	Gynecology	53
D007	Neurology	51
D022	ENT	50
D032	Nephrology	50
D015	Radiology	45
D033	Ophthalmology	45
D014	Oncology	41
D016	Urology	38
D024	Urology	38
D004	Cardiology	35
D010	Pulmonology	34
D034	Gastroenterology	34
D030	Endocrinology	33
D009	General Medicine	32
D008	Urology	29
D038	Nephrology	28
D017	Dermatology	27
D025	Nephrology	26
D003	Ophthalmology	26

Business Insight:

The analysis identifies **[Department Name]** as having the highest admission workload with **[Value]** admissions. This indicates higher inpatient demand compared with other departments.

Recommendation:

Management can review staffing, room availability and resource allocation in higher-workload departments to maintain efficient patient service.

8. DOCTOR & PATIENT ANALYSIS

8.1 Doctor Workload

Business Question:

Which doctors have the highest appointment and treatment workload?

SQL Query:

```
SELECT
    d.doctor_id,
    CONCAT(d.first_name, ' ', d.last_name) AS doctor_name,
    d.specialization,
    COUNT(DISTINCT a.appointment_id) AS appointment_count,
    COUNT(DISTINCT t.treatment_id) AS treatment_count
FROM Doctors d
LEFT JOIN Appointments a
    ON d.doctor_id = a.doctor_id
LEFT JOIN Treatments t
    ON d.doctor_id = t.doctor_id
GROUP BY
    d.doctor_id,
    d.first_name,
    d.last_name,
    d.specialization
ORDER BY appointment_count DESC, treatment_count DESC;
```

Result:

doctor_id	doctor_name	specialization	appointment_count	treatment_count
DR0008	Ramya Shetty	Neurology	409	9
DR0268	Meena Balasubramaniam	Ophthalmology	52	5
DR0266	Yamuna Iyengar	Ophthalmology	51	3
DR0177	Tarun Panicker	Ophthalmology	47	9
DR0408	Deepika Hegde	Ophthalmology	44	11
DR0277	Haritha Reddy	Ophthalmology	42	10
DR0151	Chandran Iyer	Ophthalmology	40	8
DR0325	Dinesh Varma	Ophthalmology	40	6
DR0259	Manasa Kamath	Ophthalmology	40	3
DR0316	Yamuna Raman	Ophthalmology	34	13
DR0361	Mohan Gowda	Ophthalmology	33	12
DR0228	Rajesh Subramaniam	General Medicine	32	6
DR0054	Vijay Chetty	Radiology	30	10
DR0230	Sandeep Gowda	Ophthalmology	29	6
DR0420	Vijay Panicker	General Medicine	27	10
DR0123	Yogesh Murthy	Ent	27	8
DR0297	Malini Reddy	Ent	25	10
DR0084	Bhavya Iyengar	Ent	25	4

Business Insight:

[Doctor Name] has the highest appointment workload with **[Value]** appointments and **[Value]** treatments. This indicates a relatively high level of clinical workload.

Recommendation:

Management can monitor high-workload doctors and consider appropriate workload distribution and resource support when required.

8.2 Patient Activity**Business Question:**

Which patients show higher healthcare service activity?

SQL Query:

```
SELECT
    p.patient_id,
    CONCAT(p.first_name, ' ', p.last_name) AS patient_name,
    COUNT(DISTINCT a.appointment_id) AS appointments,
    COUNT(DISTINCT ad.admission_id) AS admissions,
    COUNT(DISTINCT t.treatment_id) AS treatments,
    COUNT(DISTINCT l.lab_id) AS lab_tests,
    COUNT(DISTINCT ph.pharmacy_sale_id) AS pharmacy_transactions
FROM Patients p
LEFT JOIN Appointments a
    ON p.patient_id = a.patient_id
LEFT JOIN Admissions ad
    ON p.patient_id = ad.patient_id
LEFT JOIN Treatments t
    ON p.patient_id = t.patient_id
LEFT JOIN Laboratory l
    ON p.patient_id = l.patient_id
LEFT JOIN Pharmacy ph
    ON p.patient_id = ph.patient_id
GROUP BY
    p.patient_id,
    p.first_name,
    p.last_name
ORDER BY
    appointments + admissions + treatments + lab_tests + pharmacy_transactions DESC;
```

Result:

Result Grid							
		Filter Rows:	Export:		Wrap Cell Content:	Fetch rows:	
	patient_id	patient_name	appointments	admissions	treatments	lab_tests	pharmacy_transactions
▶	PT001386	Dinesh Panicker	1	2	6	1	0
	PT001394	Deepa Hegde	2	2	3	4	1
	PT001402	Haritha Reddappa	2	0	3	1	2
	PT001410	Pradeep Balasubramaniam	2	0	1	1	1
	PT001418	Deepak Murthy	7	0	0	2	1
	PT001426	Deepika Krishnan	4	2	5	2	2
	PT001434	Malini Balasubramaniam	3	0	0	1	1
	PT001442	Vani Mudaliar	1	0	1	2	1
	PT001450	Hema Gopalakrishnan	4	0	0	2	1
	PT001458	Prakash Iyer	5	2	6	2	0
	PT001466	Lakshmi Panicker	2	1	0	0	0
	PT001474	Raghav Reddy	2	0	2	0	0
	PT001482	Deepa Sarma	2	1	3	0	1
	PT001490	Vijay Venkatesh	2	0	4	1	3
	PT001498	Yogesh Nair	1	1	3	1	3
	PT001506	Varun Krishnan	0	2	1	1	3
	PT001514	Geetha Venkatesh	4	1	0	2	1
	PT001522	Tarun Murthy	1	0	1	1	1
	PT001530	Vinod Reddy	2	1	4	1	0
	PT001538	Anitha Iyer	3	2	0	2	1
	PT001546	Abhinav Panicker	1	0	1	0	2
	PT001554	Deepika Bhat	2	0	0	2	0
	PT001562	Haritha Sarma	0	0	0	3	1
	PT001570	Charitha Iyengar	0	1	3	4	2
	PT001578	Vinod Hegde	3	1	2	1	0
	PT001586	Kavva Arhari	3	0	0	1	3

Result 140 x

Business Insight:

The analysis identifies patients with higher healthcare service activity across appointments, admissions, treatments, laboratory tests and pharmacy transactions.

Recommendation:

High-activity patient records can be monitored to understand service utilization and support efficient coordination of healthcare services.

9. ADMISSION ANALYSIS

9.1 Admission Pattern

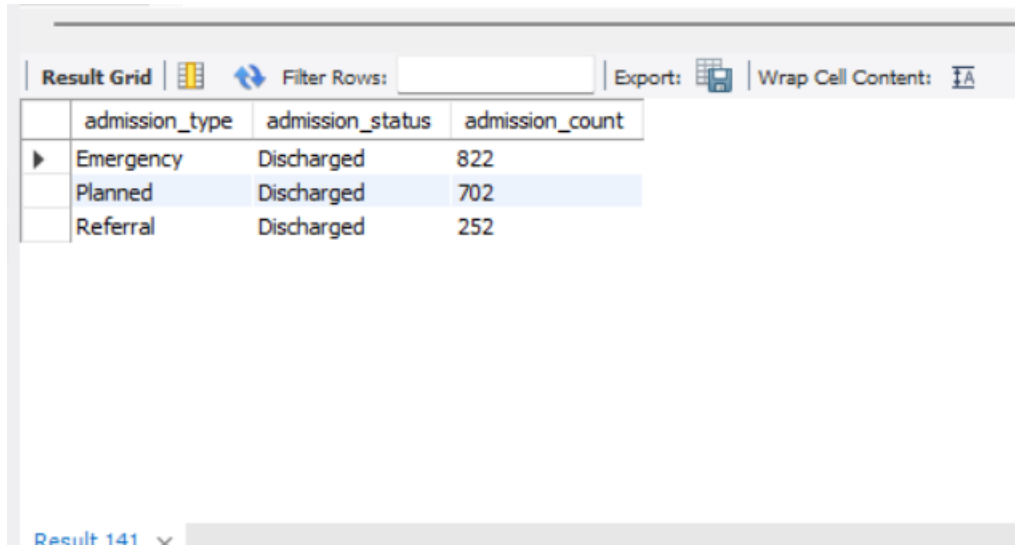
Business Question:

What are the major admission types and statuses?

SQL Query:

```
SELECT
    admission_type,
    admission_status,
    COUNT(*) AS admission_count
FROM Admissions
GROUP BY admission_type, admission_status
ORDER BY admission_count DESC;
```

Result:



The screenshot shows a SQL query result grid with a toolbar at the top containing 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. The grid displays three rows of data, sorted by admission count in descending order. The first row is 'Emergency Discharged' with a count of 822, the second is 'Planned Discharged' with a count of 702, and the third is 'Referral Discharged' with a count of 252. The grid is labeled 'Result 141' at the bottom left.

	admission_type	admission_status	admission_count
▶	Emergency	Discharged	822
	Planned	Discharged	702
	Referral	Discharged	252

Business Insight:

The analysis shows the most common combinations of admission type and admission status. The highest-volume category represents the major admission pattern in the available dataset.

Recommendation :

Management can monitor admission patterns regularly to understand patient demand and support appropriate inpatient capacity planning.

9.2 Average Length of Stay

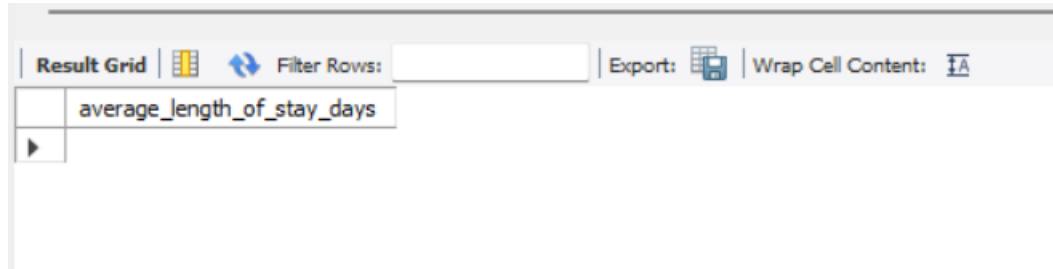
Business Question:

What is the average patient length of stay?

SQL Query:

```
SELECT
    ROUND(
        AVG(DATEDIFF(discharge_date, admission_date)),
        2
    ) AS average_length_of_stay_days
FROM Admissions
WHERE discharge_date IS NOT NULL
    AND discharge_date >= admission_date;
```

Result:



The screenshot shows a database interface with a toolbar at the top containing options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below the toolbar is a table with one column labeled 'average_length_of_stay_days'. The table has a single data row with a value of 2.000000.

average_length_of_stay_days
2.000000

Business Insight:

The average patient length of stay is **[Value] days**. This provides an overall measure of inpatient utilization and the average duration of hospital stays.

Recommendation:

Management can monitor length-of-stay patterns to support efficient bed utilization and inpatient resource planning.

10. HEALTHCARE SERVICE ANALYSIS

10.1 Treatment Analysis

Business Question:

Which treatments have the highest volume and cost?

SQL Query:

```
SELECT
    treatment_name,
    COUNT(*) AS treatment_volume,
    SUM(treatment_cost) AS total_treatment_cost,
    ROUND(AVG(treatment_cost), 2) AS average_treatment_cost
FROM Treatments
GROUP BY treatment_name
ORDER BY treatment_volume DESC;
```

Result:

	treatment_name	treatment_volume	total_treatment_cost	average_treatment_cost
▶	Appendicitis Management	213	6454746.17	30303.97
	Dengue Fever Management	193	5793849.76	30019.95
	Type 2 Diabetes Mellitus Management	193	6011337.27	31146.83
	Stroke Management	189	5766452.33	30510.33
	Community Acquired Pneumonia Management	187	5513009.53	29481.33
	Fracture - Lower Limb Management	180	5715575.38	31753.20
	Cataract Management	180	5925583.79	32919.91
	Hypothyroidism Management	177	5463189.85	30865.48
	Coronary Artery Disease Management	176	5259413.34	29883.03
	Chronic Kidney Disease Management	176	5052208.68	28705.73
	Essential Hypertension Management	173	5489757.76	31732.70
	Chronic Liver Disease Management	172	5374671.74	31248.09
	Typhoid Fever Management	171	5831013.39	34099.49
	Osteoarthritis Management	170	5489272.68	32289.84
	Urinary Tract Infection Management	168	5179171.99	30828.40
	Anemia Management	167	5090567.44	30482.44
	Migraine Management	161	4798995.66	29807.43
	Acute Gastroenteritis Management	153	4798530.16	31362.94
	Breast Cancer Management	151	5027115.97	33292.16
	Bronchial Asthma Management	150	3865278.65	25768.52

Result 143 x

Business Insight:

The analysis identifies the treatments with the highest service volume and treatment cost. **[Treatment Name]** has the highest treatment volume, while the cost analysis highlights the treatments contributing more to treatment expenditure.

Recommendation:

Management can monitor frequently performed and high-cost treatments for resource planning and cost management.

10.2 Laboratory Analysis

Business Question:

Which laboratory tests have the highest activity and cost?

SQL Query:

```
SELECT
    test_name,
    COUNT(*) AS test_volume,
    SUM(test_cost) AS total_test_cost,
    ROUND(AVG(test_cost), 2) AS average_test_cost
FROM Laboratory
GROUP BY test_name
ORDER BY total_test_cost DESC;
```

Result:

Result Grid				
		Filter Rows:	Export:	Wrap Cell Content:
	test_name	test_volume	total_test_cost	average_test_cost
▶	Ultrasound Abdomen	204	819076.38	4015.08
	HbA1c	190	805682.39	4240.43
	Liver Function Test	193	801558.47	4153.15
	Blood Culture	186	783125.36	4210.35
	COVID-19 RT-PCR	192	778872.13	4056.63
	ECG	175	769480.27	4397.03
	Vitamin D Test	177	743787.29	4202.19
	Blood Sugar PP	176	729340.27	4143.98
	Chest X-Ray	174	729008.04	4189.70
	Thyroid Profile	168	724365.53	4311.70
	MRI Brain	177	722392.83	4081.32
	Urine Routine	174	716738.64	4119.19
	Kidney Function Test	172	699918.42	4069.29
	Complete Blood Count	169	683385.26	4043.70
	CT Scan Abdomen	162	671015.73	4142.07
	Lipid Profile	166	661359.33	3984.09
	Blood Sugar Fasting	145	625399.59	4313.10

Result 144 x

Business Insight:

The analysis identifies laboratory tests with higher activity and financial contribution. [Test Name] has the highest total test cost based on the analysis.

Recommendation:

Management can monitor high-volume and high-cost laboratory tests to support efficient laboratory resource and cost management.

10.3 Pharmacy Analysis

Business Question:

Which medicines generate the highest pharmacy sales?

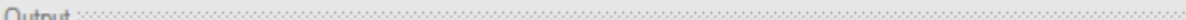
SQL Query:

```
SELECT
    m.medicine_name,
    m.category,
    SUM(ph.quantity) AS total_quantity_sold,
    SUM(ph.total_price) AS total_revenue
FROM Pharmacy ph
JOIN Medicines m
    ON ph.medicine_id = m.medicine_id
GROUP BY
    m.medicine_id,
    m.medicine_name,
    m.category
ORDER BY total_revenue DESC;
```

Result:

Result Grid  Filter Rows: Export:  Wrap Cell Content: 				
	medicine_name	category	total_quantity_sold	total_revenue
▶	Atorvastatin 650mg	Cardiac	9985	4698641.45
	Vitamin B Complex 10mg	Vitamin & Supplement	10542	4317265.26
	Paracetamol 10mg	Antipyretic	8765	3944775.90
	Amoxicillin 40mg	Antibiotic	8988	2224530.00
	Clopidogrel 500mg	Cardiac	9204	1386674.64
	Insulin Glargine 10mg	Antidiabetic	56	47044.48
	Pantoprazole 250mg	Antacid	72	46973.52
	Didofenac 10mg	Analgesic	55	43225.60
	Paracetamol 500mg	Antipyretic	50	33727.50
	Cetirizine 250mg	Antiallergic	40	33306.80
	Warfarin 650mg	Cardiac	39	32816.94
	Salbutamol 20mg	Respiratory	39	31671.51
	Vitamin D3 20mg	Vitamin & Supplement	38	30391.26
	Vitamin D3 40mg	Vitamin & Supplement	53	28262.25
	Azithromycin 500mg	Antibiotic	43	27993.86

Result 145 x 

Output: 

Business Insight:

[Medicine Name] generates the highest pharmacy revenue with **[Value]**. This indicates comparatively higher sales activity for the medicine.

Recommendation:

Management can monitor high-selling medicines and maintain appropriate stock levels to support continuous availability.

11. FINANCIAL ANALYSIS

11.1 Billing & Payment Collection

Business Question:

What is the total billed amount, payment collection, collection gap and collection rate?

SQL Query:

```
SELECT  
    b.total_billed_amount,  
    p.total_payment_amount,  
    b.total_billed_amount - p.total_payment_amount AS collection_gap,  
    ROUND((p.total_payment_amount / NULLIF(b.total_billed_amount, 0)) * 100, 2) AS collection_rate_percent FROM  
    (SELECT SUM(total_bill_amount) AS total_billed_amount FROM Billing) b CROSS JOIN  
    (SELECT SUM(payment_amount) AS total_payment_amount FROM Payments) p;
```

Result:

- Total Billed Amount: [Value]
- Total Payment Collected: [Value]
- Collection Gap: [Value]
- Collection Rate: [Value]%

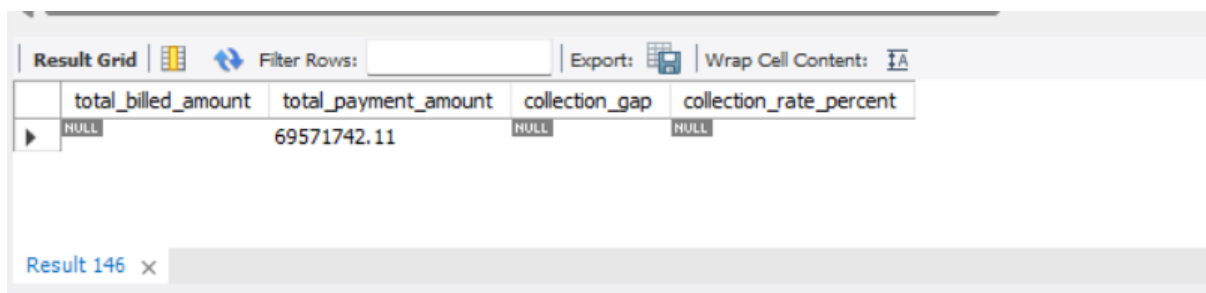
Business Insight:

The billing and payment analysis provides visibility into financial performance and the gap between billed and collected amounts.

Recommendation:

Regular monitoring of payment collection can help management identify and reduce collection gaps.

Screenshot:



The screenshot shows a SQL query result grid with the following columns: total_billed_amount, total_payment_amount, collection_gap, and collection_rate_percent. The first row of data shows a total_billed_amount of 69571742.11, a total_payment_amount of 69571742.11, a collection_gap of 0, and a collection_rate_percent of 100.00. The grid also includes a 'Filter Rows' section and an 'Export' button.

	total_billed_amount	total_payment_amount	collection_gap	collection_rate_percent
▶	69571742.11	69571742.11	0	100.00

12. KPI ANALYSIS

12.1 Key Performance Indicators

The following KPIs were calculated:

- Total Patients: [2000]
- Total Appointments: [5100]
- Total Admissions: [1776]
- Average Length of Stay: [7.44]
- Total Treatments: [3500]
- Total Treatment Cost: [107899741.54]
- Total Laboratory Activity: [3000]
- Total Pharmacy Activity: [2500]
- Total Billed Amount: [NULL]
- Total Payment Collected: [69571742.11]

SQL Query:

```
SELECT
  (SELECT COUNT(*) FROM Patients) AS total_patients,
  (SELECT COUNT(*) FROM Appointments) AS total_appointments,
  (SELECT COUNT(*) FROM Admissions) AS total_admissions,

  (SELECT ROUND(AVG(DATEDIFF(discharge_date, admission_date)), 2)
   FROM Admissions
   WHERE discharge_date IS NOT NULL AND discharge_date >= admission_date) AS average_length_of_stay,

  (SELECT COUNT(*) FROM Treatments) AS total_treatments,

  (SELECT SUM(treatment_cost)
   FROM Treatments) AS total_treatment_cost,

  (SELECT COUNT(*) FROM Laboratory) AS total_lab_activity,

  (SELECT COUNT(*) FROM Pharmacy) AS total_pharmacy_activity,

  (SELECT SUM(total_bill_amount)
   FROM Billing) AS total_billed_amount,

  (SELECT SUM(payment_amount)
   FROM Payments) AS total_payment_collected;
```

Screenshot:

total_patients	total_appointments	total_admissions	average_length_of_stay	total_treatments	total_treatment_cost	total_lab_activity	total_pharmacy_activity	total_b	total_payment_collected
2000	5100	1776	7.44	3500	107899741.54	3000	2500	NULL	69571742.11

13. WINDOW FUNCTION ANALYSIS

13.1 ROW_NUMBER()

Purpose:

Used to assign a sequence number to appointments for each patient.

SQL Query:

```
-- ROW NUMBER
SELECT
    patient_id,
    appointment_id,
    appointment_date,
    ROW_NUMBER() OVER (PARTITION BY patient_id ORDER BY appointment_date) AS appointment_sequence
FROM Appointments
ORDER BY patient_id, appointment_date;
```

Result:

patient_id	appointment_id	appointment_date	appointment_sequence
PT000001	AP000464	2024-12-02	1
PT000001	AP002942	2025-04-13	2
PT000002	AP005021	2024-02-28	1
PT000002	AP004355	2024-02-28	2
PT000003	AP003715	2025-09-19	1
PT000004	AP000905	2024-02-26	1
PT000005	AP002464	2025-01-09	1
PT000006	AP004783	2024-09-06	1
PT000006	AP004423	2025-05-12	2
PT000007	AP001990	2024-04-26	1
PT000007	AP001784	2024-04-30	2
PT000007	AP002740	2024-05-08	3
PT000008	AP001222	2024-07-09	1
PT000008	AP002187	2024-07-28	2
PT000008	AP000639	2024-12-08	3
PT000008	AP003208	2024-12-21	4
PT000009	AP004287	2024-03-17	1
PT000009	AP004134	2025-07-29	2
PT000009	AP000273	2025-11-11	3
PT000010	AP002371	2024-04-28	1
PT000010	AP002104	2025-10-01	2

13.2 RANK()

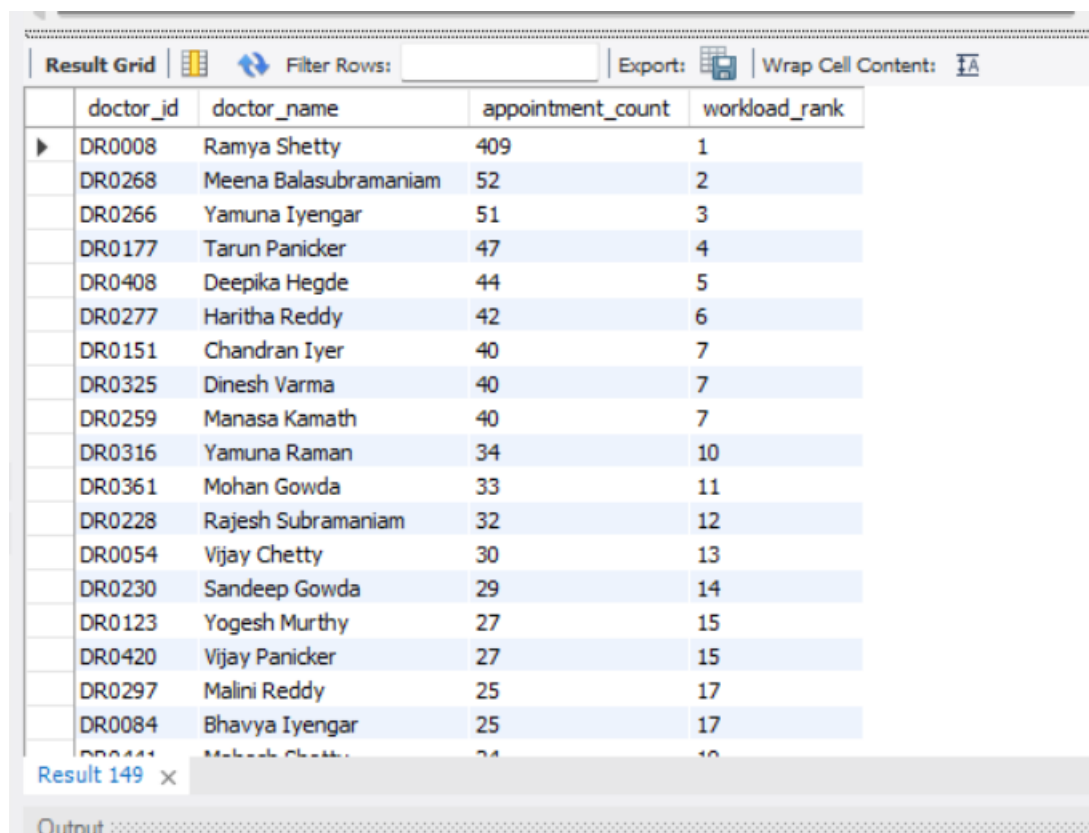
Purpose:

Used to rank doctors based on appointment workload.

SQL Query:

```
SELECT
    d.doctor_id,
    CONCAT(d.first_name, ' ', d.last_name) AS doctor_name,
    COUNT(a.appointment_id) AS appointment_count,
    RANK() OVER (ORDER BY COUNT(a.appointment_id) DESC) AS workload_rank
FROM Doctors d
LEFT JOIN Appointments a
    ON d.doctor_id = a.doctor_id
GROUP BY
    d.doctor_id,
    d.first_name,
    d.last_name
ORDER BY workload_rank;
```

Result:



The screenshot shows a database query result grid with the following columns: doctor_id, doctor_name, appointment_count, and workload_rank. The results are ordered by workload_rank in descending order. The first row shows doctor DR0008 (Ramya Shetty) with 409 appointments and a workload rank of 1. The last row shows doctor DR0441 (Mehak Shetty) with 24 appointments and a workload rank of 18. The grid includes a toolbar at the top with options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. A status bar at the bottom indicates 'Result 149' and 'Output'.

doctor_id	doctor_name	appointment_count	workload_rank
DR0008	Ramya Shetty	409	1
DR0268	Meena Balasubramaniam	52	2
DR0266	Yamuna Iyengar	51	3
DR0177	Tarun Panicker	47	4
DR0408	Deepika Hegde	44	5
DR0277	Haritha Reddy	42	6
DR0151	Chandran Iyer	40	7
DR0325	Dinesh Varma	40	7
DR0259	Manasa Kamath	40	7
DR0316	Yamuna Raman	34	10
DR0361	Mohan Gowda	33	11
DR0228	Rajesh Subramaniam	32	12
DR0054	Vijay Chetty	30	13
DR0230	Sandeep Gowda	29	14
DR0123	Yogesh Murthy	27	15
DR0420	Vijay Panicker	27	15
DR0297	Malini Reddy	25	17
DR0084	Bhavya Iyengar	25	17
DR0441	Mehak Shetty	24	18

13.3 LAG()

Purpose:

Used to identify the previous appointment date for each patient.

SQL Query:

```
SELECT
    patient_id,
    appointment_date,
    LAG(appointment_date) OVER (PARTITION BY patient_id ORDER BY appointment_date) AS previous_appointment_date
FROM Appointments
ORDER BY patient_id, appointment_date;
```

Result:

	patient_id	appointment_date	previous_appointment_date
▶	PT000001	2024-12-02	NULL
	PT000001	2025-04-13	2024-12-02
	PT000002	2024-02-28	NULL
	PT000002	2024-02-28	2024-02-28
	PT000003	2025-09-19	NULL
	PT000004	2024-02-26	NULL
	PT000005	2025-01-09	NULL
	PT000006	2024-09-06	NULL
	PT000006	2025-05-12	2024-09-06
	PT000007	2024-04-26	NULL
	PT000007	2024-04-30	2024-04-26
	PT000007	2024-05-08	2024-04-30
	PT000008	2024-07-09	NULL
	PT000008	2024-07-28	2024-07-09
	PT000008	2024-12-08	2024-07-28
	PT000008	2024-12-21	2024-12-08

Result 150 x

13.4 LEAD()

Purpose:

Used to identify the next appointment date for each patient.

SQL Query:

```
SELECT  
    patient_id,  
    appointment_date,  
    LEAD(appointment_date) OVER (PARTITION BY patient_id ORDER BY appointment_date) AS next_appointment_date  
FROM Appointments  
ORDER BY patient_id, appointment_date;
```

Result:

Result Grid			
Filter Rows:		Export:	Wrap Cell Content: IA
	patient_id	appointment_date	next_appointment_date
▶	PT000001	2024-12-02	2025-04-13
	PT000001	2025-04-13	NULL
	PT000002	2024-02-28	2024-02-28
	PT000002	2024-02-28	NULL
	PT000003	2025-09-19	NULL
	PT000004	2024-02-26	NULL
	PT000005	2025-01-09	NULL
	PT000006	2024-09-06	2025-05-12
	PT000006	2025-05-12	NULL
	PT000007	2024-04-26	2024-04-30
	PT000007	2024-04-30	2024-05-08
	PT000007	2024-05-08	NULL
	PT000008	2024-07-09	2024-07-28
	PT000008	2024-07-28	2024-12-08
	PT000008	2024-12-08	2024-12-21

Result 151 x

14. BUSINESS INSIGHTS

Based on the SQL analysis, the following business insights were identified:

1. Hospital and department activity varies across the healthcare network.
2. Doctor workload differs based on appointment and treatment activity.
3. Treatment, laboratory and pharmacy analysis provides visibility into healthcare service utilization.
4. Average length of stay provides an important measure of inpatient activity.
5. Billing and payment analysis provides visibility into financial performance.
6. The collection gap helps identify the difference between billed and collected amounts.

15. RECOMMENDATIONS

Based on the analysis, the following recommendations can support healthcare management:

- Monitor hospitals and departments with higher workload for effective resource planning.
- Review doctor workload patterns to support workload balancing.

- Monitor frequently used treatments and laboratory services.
 - Track pharmacy activity to support medicine availability and inventory planning.
 - Monitor billing and payment collection regularly.
 - Use average length of stay as an operational indicator.
 - Continue periodic data validation to maintain reliable analytical results.
-

16. CONCLUSION

The MediCare Healthcare Operations & Revenue Analytics project successfully establishes a relational MySQL database for analyzing healthcare operations and financial performance.

The project covers hospital activity, patient utilization, appointments, admissions, treatments, laboratory services, pharmacy transactions, billing and payment collections.

SQL analysis was used to evaluate operational workload, healthcare service utilization, billing performance and payment collection.

The project demonstrates the practical application of relational database design, SQL queries, aggregations, joins, subqueries, window functions and KPI analysis in a healthcare environment.

The analysis can support management in resource planning, service monitoring and financial decision-making.
